

Ablating a Hybrid Agent-Memory Architecture: A Controlled Head-to-Head of Stateless RAG, a Canonical Knowledge Graph, and Their Hybrid on a Frozen Synthetic Corpus

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Abstract

A companion study characterized a five-channel hybrid memory substrate running in production [2] and closed by naming its single most important untested claim: that the hybrid actually beats the baselines it subsumes, a comparison that report explicitly declined to run against itself (“no comparison against an external baseline ... has been run,” [2] §12; “an external-baseline experiment (plain RAG over the same corpus),” [2] §13). This study runs exactly that experiment, and turns it on its author. We build a frozen synthetic corpus of **1,583 operational facts** about a fictional field-services company, with a fully known ground-truth relation graph (414 entities, 1,498 edges, a closed vocabulary of 22 relations over 11 entity types), and put **three memory architectures behind one typed MemoryProvider interface**: (a) **stateless RAG** (chunked text, MiniLM embeddings in `sqlite-vec`, top- k only); (b) **graph-only** (the canonical-graph, closed-vocabulary posture, forward-traversal recall, no vectors); and (c) a **hybrid** that fuses facts + vectors + graph with weighted Reciprocal Rank Fusion. All three run against three seeded, frozen query sets built before any arm was executed: **150 known-item**, **30 multi-hop** (2-3 joins), and **20 distractor** queries (answer absent; the fabrication-pressure probe). Every recall is traced to an OpenTelemetry-shaped span exported into the repository. The results do not flatter the hybrid. On this clean, canonical corpus the **graph-only arm is the surprise strong performer**: it ties the field on known-item recall (Recall@5 98.7%, Recall@20 100%), wins multi-hop retrieval outright (53.3% terminal-fact success vs. the hybrid’s 40.0% and RAG’s 0.0%), and, most consequentially, is the **only arm that does not fabricate**, refusing correctly on 100% of distractors against just 10% for both RAG and the hybrid. The hybrid’s entire measured advantage reduces to ranking precision (best MRR 0.800, best Recall@1 71.3%), for which it pays the most latency (≈ 9 ms p50, $\sim 30\times$ the graph arm’s ≈ 0.3 ms) and the most storage (2,528 bytes/fact, $25\times$ the graph arm’s 101), and inherits RAG’s fabrication pressure wholesale. We state four hypotheses up front, report that two are refuted, inventory every failure, devote a section to *where the hybrid loses*, and disclose the corpus properties that make this the graph’s best case. Every number reproduces from `make eval` on the checked-in seeds.

1 Introduction

The Sheldon Dynamics memory program has, to date, produced two *characterization* studies, a fourteen-run evaluation of an operational decisioning substrate against a frozen holdout [1], and a seventeen-day in-vivo instrumentation of a production hybrid memory substrate [2]. Both study a

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single system: they measure what a shipped architecture does, in careful detail, but neither pits that architecture against the alternatives it claims to improve on. [2] is explicit and repeated about this being its central limitation. Its threats-to-validity section lists, eighth and last, “no comparison against an external baseline (hosted memory API or plain RAG over raw text) has been run,” and its recommended-next-steps section ends with “(5) an external-baseline experiment (plain RAG over the same corpus) using the E1 harness unchanged.”

This report is that experiment, deliberately generalized from “plain RAG vs. the hybrid” to a three-way ablation, and deliberately pointed back at the author’s own design assumptions. The motivating question is narrow and falsifiable: *when you strip a production hybrid memory down to its constituent postures, pure vectors, pure canonical graph, and run all three head-to-head on one frozen corpus with one frozen eval harness, does the hybrid actually earn its complexity, and where does it not?*

The scientific value of the study is precisely that it can embarrass its author, and partly does. A hybrid that swept every metric would be a red flag: it would suggest a benchmark tuned to the conclusion. What follows instead is an ablation in which the simplest structured arm wins most axes on this corpus, the hybrid’s advantage is real but narrow and expensive, and the corpus itself is disclosed as the graph’s most favorable case. That combination, a genuine cost paid, honestly reported, is the point. Synthetic is a feature, not a concession: unlike a production store, a synthetic corpus with a generated ground-truth graph is shareable, re-runnable, and gives every query a machine-checkable answer, including the distractors whose correct answer is “not in memory.”

2 Hypotheses

Stated before results, judged against them in §11.

- **H1: Vectors win known-item.** The vector-bearing arms (RAG, hybrid) will lead known-item retrieval (Recall@5, MRR); the closed-vocabulary graph, lacking dense semantics, will trail on lexically-paraphrased single-fact queries.
- **H2: Structure wins multi-hop.** Graph and hybrid will beat RAG on multi-hop retrieval by a wide margin, because joins require traversal that top- k vector recall cannot perform.
- **H3: Refusal is ordered graph > hybrid > RAG.** Graph-only will refuse most correctly on distractors (absence from a canonical graph is an unambiguous signal), RAG will fabricate most, and the hybrid will land *between* the two.
- **H4: The hybrid does not dominate.** It will not win every metric; it will pay measurable costs in latency, storage-per-fact, and at least one quality axis relative to the specialized arms.

3 The three arms

All three implement one typed interface, `MemoryProvider` (`arms/base.py`): `build(facts)` indexes the frozen corpus once; `recall(query, k)` returns a ranked list of facts plus a single boolean, `refused`, the arm’s own decision that it has no confident answer. The harness derives every metric from that response and the wall-clock time of the call, so no arm sees ground truth or can special-case the evaluation.

(a) **Stateless RAG (rag)**. Each corpus fact is one retrieval chunk (the facts are sentence-sized). Facts are embedded with `all-MiniLM-L6-v2` (384-d, CPU, L2-normalized) and stored in a `sqlite-vec (vec0)` index; recall is exact-cosine top- k . There is no graph, no entity linking, no state between queries. A single cosine floor (0.40, matching the source system’s semantic threshold [2]) is both the relevance cutoff and the entire refusal rule: if the best match is below the floor, the arm reports “not in memory.” That one threshold is RAG’s only defense against fabrication.

(b) **Graph-only (graph)**. The corpus’s ground-truth relation graph is loaded into a forward (subject→object) adjacency structure. A query is linked to seed entities by a shared, purely-lexical linker (id patterns like `W0-10047`, plus whole display-name matches on word boundaries); recall is a bounded forward breadth-first traversal that collects each reached node’s *outgoing* facts, ranked by hop distance then token-Jaccard overlap. No vectors. Forward-only traversal is deliberate: every relation points outward from an entity to its resources (a job to its equipment, a site to its owner), so it follows every reasoning chain while avoiding the reverse fan-in explosion through hub nodes, the same hub-suppression instinct [2] encodes, expressed here as edge directionality. The arm refuses exactly when the query links no entity.

(c) **Hybrid (hybrid)**. A ported slice of the five-channel design [2], facts + vectors + graph, with three grounded channels: a BM25 lexical index over fact text, MiniLM cosine (shared embedder with RAG), and the graph traversal (shared engine with the graph arm). The three channel orderings are combined by **weighted Reciprocal Rank Fusion** [3], with the graph channel up-weighted 2:1 because it is the precision-oriented channel and the only one that can reach a multi-hop answer. This is the parameter-free fusion [2] §13 named as its own top recommended improvement. The refusal rule is deliberately permissive, the hybrid refuses only when *both* the vector top is below the floor *and* the graph links no seed; the BM25 channel is a recall booster, never a grounding signal. §8 reports the consequence.

3.1 Design constants

Group	Constant	Value
Embedder	model / dim	all-MiniLM-L6-v2 / 384
Vector	store / search	<code>sqlite-vec vec0</code> / exact cosine
Vector	cosine floor (relevance + RAG refusal)	0.40
Graph	traversal / max hops	forward BFS / 3
Graph	candidate rank key	(hop asc, Jaccard desc, id)
Hybrid	fusion	weighted RRF, $k_0 = 60$
Hybrid	channel weights (vector : facts : graph)	1 : 1 : 2
Hybrid	grounded-refusal rule	vector < 0.40 and no graph seed
Eval	known-item depth / cutoffs	20 / {1, 5, 20}
Eval	multi-hop judged at	top-10
Seeds	corpus / query	42 / 1234

Table 1: Design constants. Fixed and committed; nothing is tuned per query or run.

4 The corpus

`corpus/generate.py` deterministically builds **Meridian Field Services**, a fictional industrial field-services company, from a single seeded RNG. Population sizes are fixed constants (seed-independent), so the corpus size is stable across machines; the seed controls only the random wiring and phrasing. Every fact is derived from exactly one graph edge or node attribute, so every fact has a known subject, relation, object, and set of linked entities.

Layer	Count	Notes
Facts	1,583	one NL statement per edge/attribute
Entities	414	60 people, 45 equipment, 20 sites, 12 clients, 10 vendors, 6 teams, 6 certs, 150 jobs, 40 incidents, + city/date nodes
Edges	1,498	directed, typed
Relations	22	closed vocabulary (REPORTS_TO, USES, MAINTAINED_BY, ...)
Entity types	11	closed vocabulary

Table 2: Substrate inventory of the frozen corpus (seed 42).

Fact composition mirrors a real operational store: **jobs dominate row count** (834 facts, 53%) the way email exhaust dominates the production store in [2], while org, equipment, incident, and site facts make up the rest. This is deliberate, it lets known-item queries stratify across categories while the bulk sits in one operational category, reproducing the “row count is not retrieval demand” property [2] observed in production. The generator writes `facts.jsonl`, `graph.json`, and a `manifest.json` carrying SHA-256 checksums of both; the committed corpus is verified against a fresh regeneration in the test suite.

5 The query sets

All three query sets are built by `harness/build_queries.py` from the corpus and its ground-truth graph, **before any arm is run**, and committed with a seed and checksums.

- **Known-item (150).** A natural question whose answer is one specific fact, generated from functional relations across all five categories and phrased with paraphrase (“Who is X’s supervisor?” for a `REPORTS_TO` fact). Ground truth: the gold fact id and its answer entity.
- **Multi-hop (30).** A question whose answer requires joining 2-3 facts (22 two-hop, 8 three-hop), from ten templates over forward chains (e.g. *job* → *USES* → *equipment* → *MAINTAINED_BY* → *vendor*). Ground truth: the terminal fact that yields the answer entity, the full chain, and the answer entity. Every chain is verified forward-reachable from a query-mentioned seed.
- **Distractor (20).** A well-formed question about an entity that is **not** in the corpus. The correct behavior is refusal. The test suite asserts each distractor links to zero real entities, a clean absence probe.

The distractors are the sharpest instrument: phrased identically to answerable known-item queries but naming an absent entity, they are lexically and semantically close to real facts. That is what makes them a fabrication-pressure test: a dense retriever will find *something* similar and must decide whether to answer.

6 Method

Three instruments, all deterministic and seeded. (1) **Retrieval replay**: each arm is built once over the 1,583-fact corpus, then answers all 200 queries; known-item scored at cutoffs {1,5,20} from a depth-20 recall, multi-hop at top-10, distractors on the **refused** flag. (2) **Latency microbenchmark**: every **recall** call is wall-clock timed after a per-arm warmup; p50/p95 reported over all 200 queries and host-dependent, as in [2]. (3) **Tracing**: every recall is emitted as an OpenTelemetry-shaped span and exported to **traces/<arm>/** as OTLP JSON, shipping a standard-tooling artifact and closing the “own-telemetry-only” gap [2] flagged, as a side effect. The headline numbers use **all-MiniLM-L6-v2** on CPU (deterministic to float32 given the pinned version); a dependency-free hashing embedder (**AMA_EMBEDDER=hash**) lets the harness run fully offline, and because all three arms share whichever embedder is selected, the architectural comparison is valid under either.

7 Results

All numbers are produced by **make eval** with the MiniLM embedder and are the exact contents of **results/tables.md** (auto-generated; not hand-edited).

7.1 Known-item retrieval

Arm	Recall@1	Recall@5	Recall@20	MRR
rag	55.3%	68.7%	75.3%	0.610
graph	62.0%	98.7%	100.0%	0.750
hybrid	71.3%	98.0%	100.0%	0.800

Table 3: Known-item retrieval (150 queries).

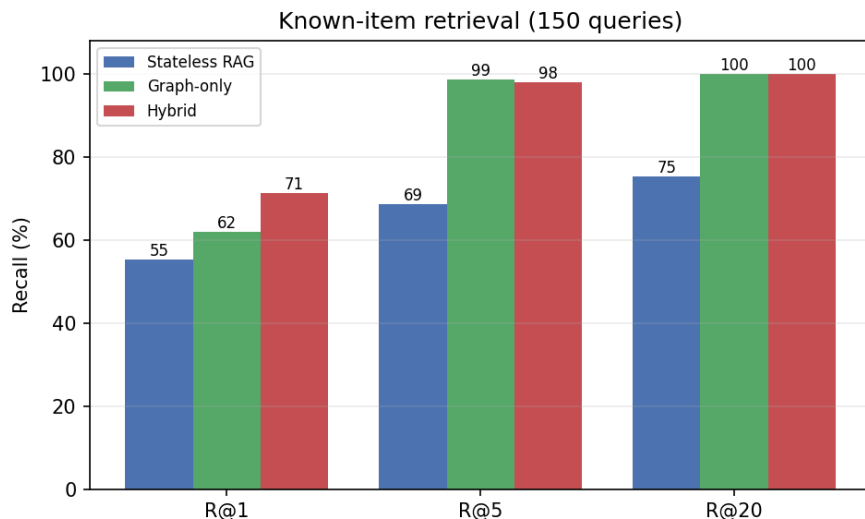


Figure 1: Known-item retrieval. The channels err on different queries.

The first surprise is at the top of the table: **pure RAG is the weakest known-item arm**, and by a wide margin at depth (Recall@20 of 75.3% means one gold fact in four never surfaces

in the top 20). The cause is directional confusion in a duplicate-rich store: a query like “Who is Dana Reyes’s supervisor?” is embedded close to every “ X reports to Dana Reyes” fact where Dana is the *object*, and those crowd out the single fact where she is the subject. The graph arm has no such problem, it links the subject and returns that subject’s small set of outgoing facts (Recall@20 = 100%). The hybrid wins ranking precision (Recall@1 71.3%, MRR 0.800). **H1 is refuted:** on an entity-anchored corpus, structure beats pure semantics even for single-fact retrieval.

7.2 Multi-hop retrieval

Arm	Terminal-fact success (top-10)	Answer-entity recall	Chain coverage
rag	0.0%	3.3%	5.0%
graph	53.3%	53.3%	81.1%
hybrid	40.0%	40.0%	74.4%

Table 4: Multi-hop retrieval (30 queries).

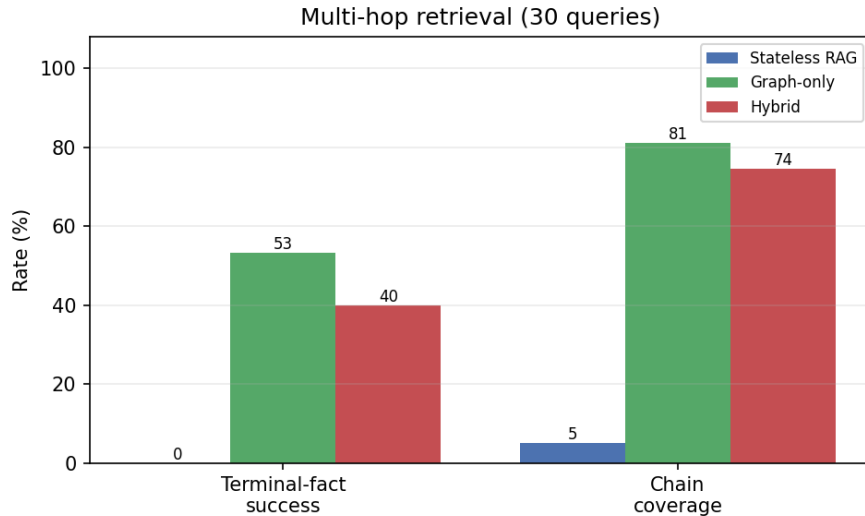


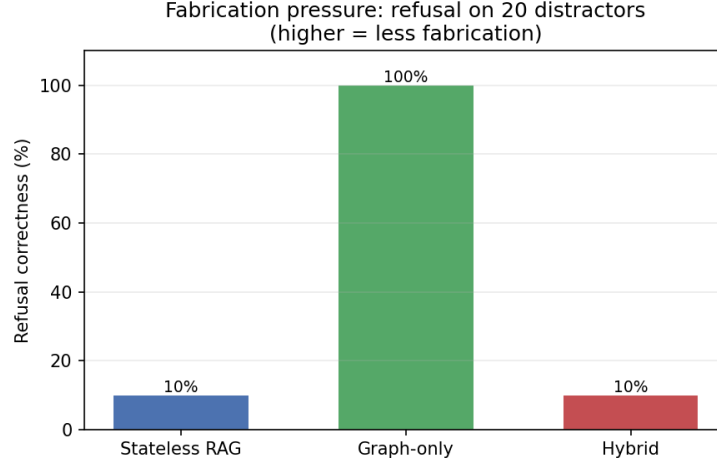
Figure 2: Multi-hop retrieval. Graph recalls the chain; the terminal ranks are the gap.

H2 is confirmed, sharply. RAG cannot join: it retrieves facts surface-similar to the question, which name the anchor entity but never the terminal answer, so it succeeds on zero of thirty. Graph and hybrid traverse. The nuance is the gap between *chain coverage* and *terminal success*: the graph arm surfaces 81% of all chain facts but floats the specific terminal fact into the top 10 only 53% of the time, it recalls the reasoning chain (the hard part) but a lexical ranker does not prioritize the answer within it. The hybrid trails the graph on all three measures despite containing the graph channel; §8 explains why.

7.3 Refusal and fabrication

This is the study’s most consequential table. The graph arm **never fabricates**: an absent entity links nothing, so it refuses correctly on all 20 distractors while never over-refusing any of the 180 answerable queries. RAG fabricates on 90% of distractors, because they are phrased like real

Arm	Refusal correctness (20 distractors)	Over-refusal (180 answerable)
rag	10.0%	0.0%
graph	100.0%	0.0%
hybrid	10.0%	0.0%

Table 5: Refusal & fabrication.**Figure 3:** Fabrication pressure: refusal on 20 distractors (higher = less fabrication).

queries, the nearest fact clears the 0.40 cosine floor and the arm answers with confident nonsense. And the hybrid, despite carrying the graph channel that would have refused, **fabricates exactly as often as RAG. H3 is refuted:** the hybrid is not intermediate; it is pinned to RAG’s fabrication rate.

7.4 Cost: latency and storage

Arm	Latency p50 (ms)	Latency p95 (ms)	Index bytes	Bytes/fact
rag	7.68	10.07	3,244,032	2,049
graph	0.30	0.52	159,744	101
hybrid	8.98	11.22	4,001,792	2,528

Table 6: Cost. Latency is the one host- and run-dependent row (a single reference run); index bytes and every retrieval metric are exact and stable.

The graph arm is $\sim 30\times$ **faster** at the median than the hybrid (≈ 0.30 ms vs. ≈ 9 ms) and uses $25\times$ **less storage per fact** (101 vs. 2,528 bytes), because it holds no embeddings and touches no embedder on the hot path. The two vector arms are dominated by the ~ 8 ms MiniLM query embedding; the hybrid adds BM25 and graph work on top. **H4 is confirmed.**

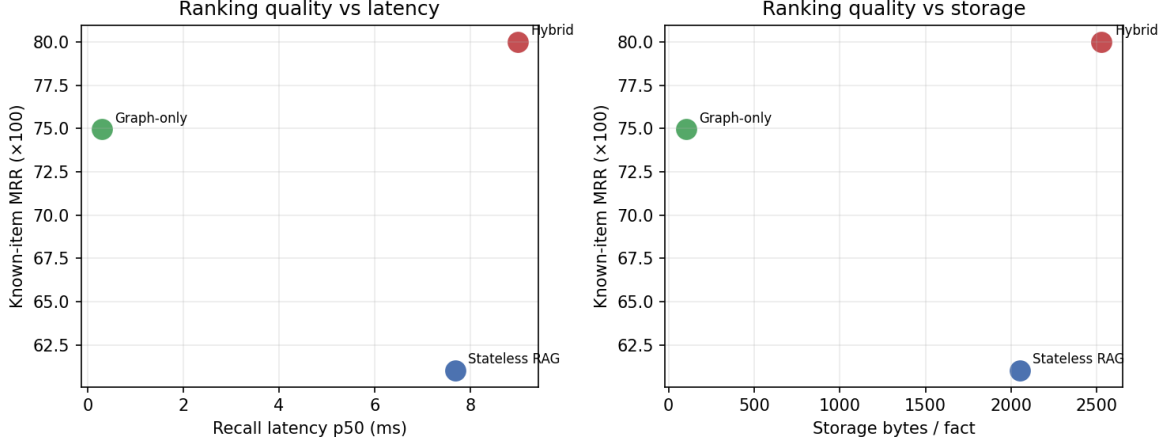


Figure 4: Ranking quality (MRR) versus the two costs, latency and storage.

8 Where the hybrid loses

A hybrid that won everything would be a benchmark artifact. This one loses three ways, two of them by inheriting the failure modes of the channels it fuses, and the honest accounting is that its *only* wins on this corpus are ranking-precision (Recall@1, MRR).

1. It re-inherits RAG’s fabrication (the expensive loss). The hybrid contains the very channel, graph linking, whose absence signal drives the graph arm to a perfect 100% on distractors. It still fabricates at RAG’s rate (10%) because its grounding rule is a permissive OR: it answers if *either* the vector channel clears the floor *or* the graph links a seed. On a distractor the graph correctly abstains, but the vector channel clears the floor and the OR lets it speak. This is the general hazard of naive multi-channel grounding: **union-of-evidence maximizes recall and minimizes refusal**, and refusal is where fabrication lives. A graph-*veto* rule would recover the 100%, at the cost of over-refusing vector-answerable queries the graph cannot link. That trade is real and unmeasured here; it is the single highest-leverage follow-up (§12).

2. Fusion dilutes the graph channel on multi-hop (the subtle loss). The hybrid trails graph-only on every multi-hop measure (40.0% vs. 53.3% terminal success) *even with the graph channel up-weighted 2:1*. Reciprocal Rank Fusion rewards cross-channel agreement, but a multi-hop terminal fact is found by exactly one channel, the graph, while the two high-recall channels agree on shallower, surface-similar facts and outvote it. Weighting the graph channel higher narrows the gap (unweighted RRF scored 13.3%; the 2:1 weight lifts it to 40.0%) but does not close it. Fusion, the mechanism that buys the hybrid its known-item ranking win, is the same mechanism that costs it multi-hop recall.

3. It is the most expensive arm on every cost axis (the unavoidable loss). Three materialized indices (vectors + BM25 + graph) make it the largest store at 2,528 bytes/fact, and three channels plus a query embedding make it the slowest at ≈ 9 ms p50. It is strictly Pareto-dominated by the graph arm on latency, storage, multi-hop, and refusal simultaneously; it beats the graph arm only on MRR (0.800 vs. 0.750) and Recall@1 (71.3% vs. 62.0%). The honest one-line valuation: **it buys ~ 9 points of Recall@1 and 0.05 of MRR for $\sim 30\times$ the latency, $25\times$**

the storage, and a 90-point refusal regression. What rescues the hybrid is not visible in this corpus at all (§10), and naming that is the price of credibility.

9 Failure inventory

- **RAG, directional confusion (known-item).** 24.7% of gold facts never reach RAG’s top-20; the dominant mode is subject/object inversion in symmetric-looking relations.
- **RAG, no-join (multi-hop).** 30/30 terminal misses (5.0% chain coverage); no mechanism to follow an edge.
- **RAG & hybrid, confident fabrication (distractor).** 18/20 distractors answered; the nearest real fact clears the 0.40 floor and neither arm has a structural absence signal to override it.
- **Graph, terminal-rank burial (multi-hop).** 14/30 terminal misses despite 81% chain coverage; hop-ascending order front-loads the anchor’s shallow facts and Jaccard cannot distinguish the paraphrased terminal. All 8 three-hop queries fail this way.
- **Graph, paraphrase ranking (known-item MRR).** MRR 0.750 trails the hybrid’s 0.800 when the query shares no salient token with the gold fact.
- **Hybrid, fusion dilution.** See §8.2.

10 Threats to validity

1. **The corpus is the graph’s best case, and we say so first.** Meridian is synthetic, fully canonical (one clean id per entity, a closed relation vocabulary, zero extraction noise), and its queries are entity-anchored. That is precisely the regime in which entity linking and traversal dominate and dense semantics are least needed. Real operational memory, the production setting [2] the hybrid was built for, is the opposite: noisy distilled text, no clean ids in the user’s phrasing, heavy paraphrase and coreference. The single most likely way these results *fail to generalize* is that they understate the vector and hybrid arms in exactly the conditions those arms exist to handle.
2. **Known-item queries name the anchor entity,** handing the graph and hybrid arms a free, reliable seed, the primary driver of their known-item lead.
3. **Multi-hop success is a strict top-10 terminal-fact bar,** undercounting the graph arm’s 81% chain coverage.
4. **Single embedder, single host.** A stronger embedder would lift the vector arms uniformly; latency is from one 2-vCPU-class host.
5. **The refusal thresholds are pre-committed, not optimal.** RAG’s 0.40 floor is inherited from [2], not tuned on this corpus; we fixed it in advance to avoid post-hoc curve-fitting.
6. **The hybrid is one point in a design space.** Weighted RRF with a 2:1 graph weight and OR-grounding is a reasonable, minimal hybrid, not the optimal one; §8 names two variants that would move its numbers.
7. **No LLM in the loop.** These are retrieval metrics, not end-task answer-accuracy metrics.

11 Discussion

The hypotheses, revisited. H2 and H4 are confirmed. H1 and H3 are **refuted**: pure RAG was the *worst* known-item arm on an entity-anchored corpus, and the hybrid did not sit between graph and RAG on refusal, it was pinned to RAG’s fabrication rate. Refuting one’s own hypotheses is the intended outcome of a real ablation.

The substrate thesis, ablated. [2] argued that recall quality is governed by store structure, not model choice. This study sharpens it: on a clean corpus, *structure alone* (the graph arm, no model on the hot path) is competitive-to-dominant, and the dense channel’s marginal contribution is ranking precision plus the coverage the graph lacks when queries are not entity-anchored, which this corpus, by construction, does not test. The production hybrid’s justification therefore lives in the gap between this corpus and real operational text, and this study locates that gap rather than papering over it.

What the fabrication result means for hybrids generally. The most transferable finding is architectural: **fusing a high-precision abstaining channel with a high-recall answering channel under union-of-evidence grounding throws away the abstention.** The graph arm’s zero-fabrication property is a design asset a naive hybrid destroys. Any hybrid that wants both recall and calibrated refusal must make refusal a *veto* of the grounded channel, not a *vote*.

12 Recommended next steps

In order of leverage: (1) **graph-veto refusal** in the hybrid, measuring recovered distractor refusal against induced over-refusal; (2) **rank-aware fusion** that preserves a lone deep-hop graph hit; (3) a **paraphrase / descriptive-reference query set** that does not name the anchor entity, the single most important generalization test; (4) a **noisy corpus variant** (extraction noise, alias drift, coreference) moving Meridian toward production conditions; (5) an **LLM answer-accuracy layer** that converts retrieval metrics into task metrics.

13 Conclusion

Three memory architectures, one frozen 1,583-fact corpus, 200 seeded queries, one eval harness: on this clean, canonical corpus the **graph-only arm wins or ties most axes**, Recall@20 100%, multi-hop 53.3%, refusal 100%, at ≈ 0.3 ms and 101 bytes/fact, while the **hybrid’s measured advantage reduces to ranking precision** (Recall@1 71.3%, MRR 0.800), bought at $\sim 30\times$ the latency, $25\times$ the storage, and a 90-point refusal regression it inherits wholesale from its dense channel. Two of four pre-registered hypotheses are refuted. The corpus is disclosed as the graph’s most favorable case, which is exactly where the honest boundary of the claim lies: this study does not show that hybrids are unnecessary; it shows that on canonical, entity-anchored memory the specialized graph is hard to beat and cheap to run, that a naive hybrid throws away the graph’s zero-fabrication property, and that the production hybrid’s real justification lives in the messier corpus this study deliberately did not build, and names as the next one to.

Practitioner one-liner. On a clean, canonical operational corpus, a vector-free knowledge graph beats plain RAG and ties or beats a fused hybrid on almost everything that matters, recall, multi-hop, and especially refusal (100% vs. 10%), at $\sim 1/25$ th the storage and $\sim 1/30$ th the latency; the hybrid’s extra machinery buys ranking precision and, unless refusal is a veto rather than a vote, silently reinherits RAG’s fabrication.

Reproducibility. A stranger reproduces every number with `pip install -r requirements.txt` then `make eval` (or `python -m harness.run_eval`), which regenerates the frozen corpus and query sets (verifying committed checksums), builds all three arms, runs the 200 queries, and writes `results/*.csv`, `results/tables.md` (the source of Tables 1-4), `results/summary.json`, `traces/<arm>/` (OTLP spans), and the figures. All retrieval metrics are deterministic given the pinned `all-MiniLM-L6-v2`; only latency (Table 4) is host-dependent. `AMA_EMBEDDER=hash` runs the whole study offline with no model download. A 25-test suite (`make test`) fails if the committed artifacts drift from their seeds.

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